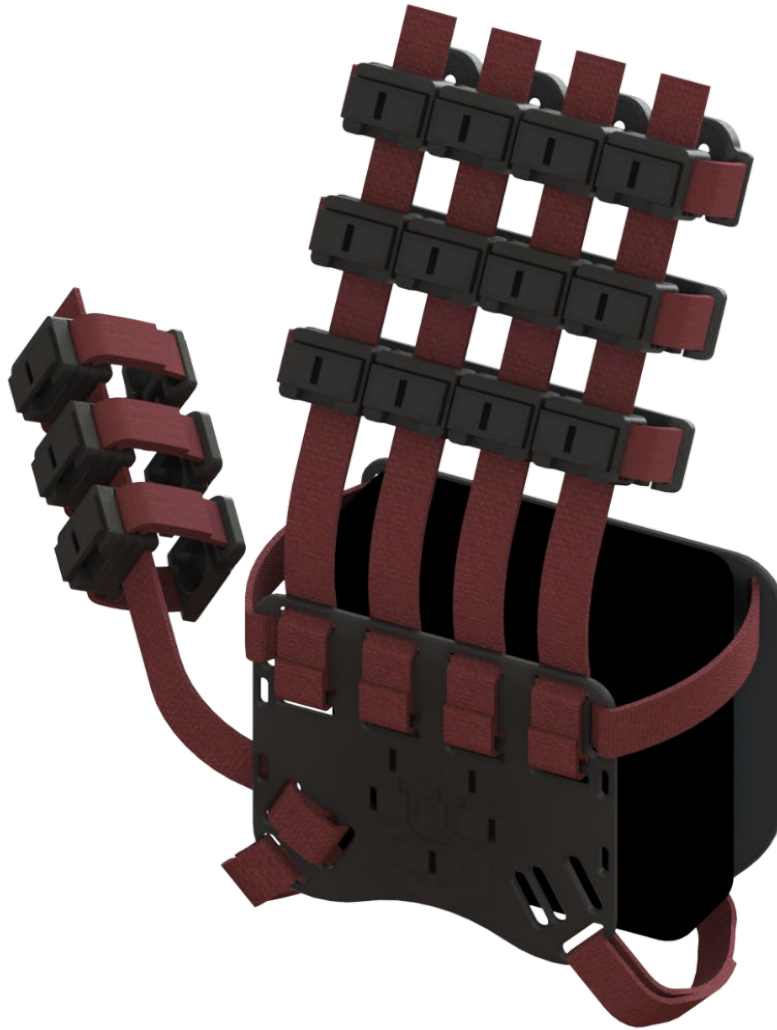




HAND SENSORY REHABILITATION FOR STROKE PATIENTS

TEAM 17

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Executive Summary

After a stroke, the sensory cortical maps within the brain can become scrambled, leading to sensory inputs becoming confused, particularly in patients' hands. For example, a patient could receive a stimulus to their thumb, but the perceived stimulus is experienced in the ring finger. These sensory impairments also cause stiffness and decreased mobility which are addressed in current therapies which focus on stretching, massage, and functional use of the hand; however, there is an intrinsic difficulty in reconnecting the affected neural pathways.

The goal of this project was to design, manufacture, and test a wearable device to be used in a new method of sensory rehabilitation for stroke patients. To facilitate this form of rehabilitation, the device implements vibration stimulus paired with visual stimulus. A total of 20 pairs of Eccentric Rotating Mass vibration motors (ERMs) and light-emitting diodes (LEDs) were implemented to provide the necessary stimuli. The ERMs and LEDs were arranged in groups of three ERM and LED pairs per finger and five pairs in the palm. For patient comfort and adaptability, the vibrational intensity, in the coupled form of frequency and amplitude, can be adjusted by the user.

The device is controlled through a Graphical User Interface (GUI) and can be operated on any personal computer. The GUI allows users to turn different motor and LED pairs on and off either individually or in groups, such as entire digit sections or the whole palm. To simplify the complexity of the electronic connections, a custom Printed Circuit Board (PCB) was designed and manufactured. For safety, the PCB and electronics are safely stored in an electrical box.

The wearable portion of the device is lightweight for user comfort and to minimize user fatigue. The device is also highly adjustable, fitting 90% of adult stroke patients, and may be used on either hand with minor adjustments. Putting the device on is relatively simple and can be accomplished in under five minutes; however, due to the adjustability of the device, assistance will often be required to attain the proper fit. The rehabilitation device and GUI meet all user needs and will be used in the clinical setting to assess the effectiveness of the paired visual and vibrational stimuli in hand sensory rehabilitation for stroke patients.



Rehabilitation device showing the electrical box (red) and the wearable portion of device consisting of 20 LEDs and vibrational motors.

Acknowledgements

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Project Overview

After a stroke, the sensory cortical maps within the brain can become scrambled which can lead to stroke patients having jumbled perception of the senses in their hands. For example, a patient could receive a stimulus to their thumb, but the perceived stimulus is experienced in the ring finger. These sensory impairments also cause stiffness and decreased mobility which are addressed in current therapies. These therapies focus on stretching, massage, and functional use of the hand; however, there is an intrinsic difficulty in reconnecting the affected neural pathways.

This project aimed to address these jumbled cortical pathways by creating a first step in physical therapy for stroke patients on their affected hand. The device created in this project will be used in future therapies to provide vibrational stimulus in different locations along the hand. Visual stimulus is paired with vibrational stimulus, allowing patients to associate the tactile feeling in their hand with the physical location of the visual stimulus. The resulting therapy techniques will aim to help patients connect the perceived location of the sensation with the true location on the hand, thereby helping to reconnect and re-establish the affected cortical maps.

To make a product that satisfied all user needs, shown in Table 1, the wearable haptic device, like a glove, was designed and manufactured with 20 vibrational motors and 20 LEDs. The motors are located across the palmar side of the hand with three per finger and five on the palm. On the dorsal side of the hand, LEDs were implemented to correspond with the locations of the motor. The glove that houses the motors, LEDs, and wiring was designed to allow for adjustability to accommodate a wide variety of hand sizes, mobility levels, and handedness while also optimizing for ease of patient use.

The wearable device connects to a Graphical User Interface (GUI) that allows the user to activate different motors individually or in groups and adjust the vibrational intensity and the duration of the applied pulses. The GUI is accessible through an executable file created with LabVIEW that can be downloaded on any computer for increased accessibility and was designed with clinicians in mind.

Table 1: User needs and their descriptions ranked by importance.

Rank	User Needs	Description
1.	Safety	Device shall be safe for the user
2.	Stimulus	Device shall provide both vibratory and visual stimulus
3.	Controllability	Motors/LED shall be controllable
4.	User Fit	Device shall fit 90% of adult stroke patients
5.	User Comfort	Device shall be comfortable to wear and operate
6.	Ease of Use	Device shall be easy to use

Design Specifications

Design specifications were derived from the user needs and are shown in Table 2. To ensure user safety, metrics 1-5 were established which set limits on the vibrational frequency and amplitude as well as ensure that there are no sharp edges or exposed wires. Metrics 3-5 establish the need for vibrational and visual stimulus. The vibrational and visual stimuli needed to be controllable which was accounted for in metrics 6-9. Lastly, the glove needed to fit 90% of adult users (metrics 11-13), be comfortable to wear and operate (metrics 10-11) and be easy to use (metrics 14-16).

Table 2: Design specifications with the corresponding units, along with the associated user need(s), as well as the target values, and the testing results.

Metric		Units	Target	Result	User Need
1 -	No exposed wiring	Binary (Yes/No)	Yes	Yes	1
2 -	No sharp edges	Binary (Yes/No)	Yes	Yes	1
3 -	Device provides visual stimulus	Binary (Yes/No)	Yes	Yes	2
4 -	Minimum vibration frequency	Hertz	> 100	150	1, 2
5 -	Maximum vibration amplitude	Gs	< 1	0.75	1, 2
6 -	Device controllable using a GUI	Binary (Yes/No)	Yes	Yes	3
7 -	Motors individually controllable	Binary (Yes/No)	Yes	Yes	3
8 -	Motors controllable in groups	Binary (Yes/No)	Yes	Yes	3
9 -	Vibration intensity controllable by user	Binary (Yes/No)	Yes	Yes	3
10 -	Comfortability to wear and operate	Ranked Level (1-5)	> 3	3.5	5
11 -	Weight (of wearable portion)	Pounds	< 2	0.32	5
12 -	Maximum finger length	Inches	> 3.5	6	4
13 -	Maximum finger strap length	Inches	> 1	2	4
14 -	Time to put device on assisted	Minutes	< 3	2.8	6
15 -	Time to put device on unassisted	Minutes	< 5	4.5	6
16 -	Ease of putting on device	Ranked Level (1-5)	> 3	3	6

Conceptual Design

Four design concepts were generated for both the wearable portion of the design as well as the control system, and then ranked. The concepts are listed below:

Wearable design:

1. Create a device with a fabric palm that houses the palm motors and use hook-and-loop straps to hold the motors and LEDs in place on the fingers.
2. Use a pre-made glove and retrofit it with motors and LEDs.

3. Create a glove-like device out of plastic. The full glove design would be 3D printed with predetermined locations for the motors and LEDs.
4. Create a table-top tablet-like device that would house the motors and LEDs on a flat plane with a physical indentation for the patient to place their hand.

Control system:

1. Generate a MATLAB GUI.
2. Generate a LabVIEW GUI.
3. Create a physical control panel with knobs and buttons.
4. Purchase a tablet and create an application.

Designs were then ranked from 1 to 3 as follows for cost, practicality, adjustability, comfort, and ease-of-use as applicable:

1. Negative expected outcome
2. Neutral expected outcome
3. Positive expected outcome

After rating each of the designs, the average ratings were calculated and used to rank the design concepts from best to worst, see Tables 3 and 4.

Table 3: Decision matrix for the design ideas ranked subjectively from 1 to 3 based on their cost, practicality, adjustability, comfort, and ease-of-use. 1 corresponding to a negative expected outcome, 2 to a neutral expected outcome, and 3 to a positive expected outcome.

Wearable Design						
Design Concepts	Cost	Practicality	Adjustability	Comfortability	Ease of Use	Average
Fabric Design	3	3	3	3	2	2.8
Premade Glove	3	2	1	3	3	2.4
Plastic Glove	1	1	2	1	2	1.4
Table-Top Design	2	2	1	1	1	1.4

Table 4: Decision matrix for the control system ideas ranked subjectively from 1 to 3 based on their cost, ease-of-use, and practicality. 1 corresponding to a negative expected outcome, 2 to a neutral expected outcome, and 3 to a positive expected outcome.

Control System				
Design Concepts	Cost	Ease of Use	Practical	Average
LabVIEW GUI	3	3	3	3
MATLAB GUI	3	2	3	2.67
Physical control panel	2	3	3	2.67
Tablet	1	2	1	1.33

From the design concept analysis, a device made up of fabric with hook-and-loop straps was deemed the most likely to succeed as it was expected to be the cheapest, most practical, most comfortable, and was not expected to be difficult to use. For control, a LabVIEW GUI ranked the best in all categories and allowed the creation of an executable that can be easily downloaded and used on any computer as opposed to MATLAB. The two optimal concepts were pursued in the final design.

Final Design

Device Design

The device was manufactured out of 3D printed parts, fabric, hook-and-loop straps, and nylon webbing straps and contains the device motors and LEDs. A solid model of the design can be seen in Figure 1 and a picture of the final device can be seen in Figure 3. The individual components in the figures are labeled with part numbers corresponding to the Bill of Materials in Appendix 3.

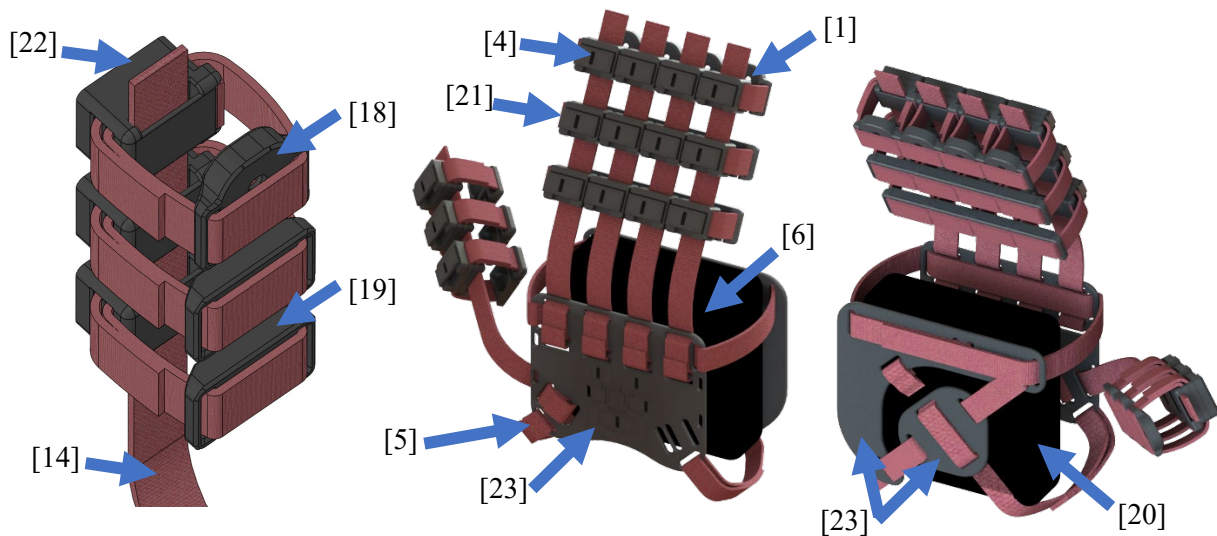


Figure 11: Solid model of the wearable device. The finger design is shown in detail (left), and the entire design is shown from the dorsal side of the hand (center) and from the palmar side of the hand (right). Component information is located within the Bill of Materials in Appendix 6.

Three distinct motor mounts were designed to account for the differences in geometry between the fingertip, the rest of the finger, and the palm. The different geometries can be seen in Figure 2. The fingertip motor mount was designed to place the motor toward the tip of the finger while the other finger mounts hold the motor in the center of the finger segment. Both finger designs include slots for the hook-and-loop straps, allowing them to be strapped to the fingers and adjusted circumferentially about the finger. The palm motor mounts were designed to be sewn or glued into the foam and fabric of the palm section. The motor leads were hot glued into the wire routing channels within the mounts to provide strain relief and improve user safety.

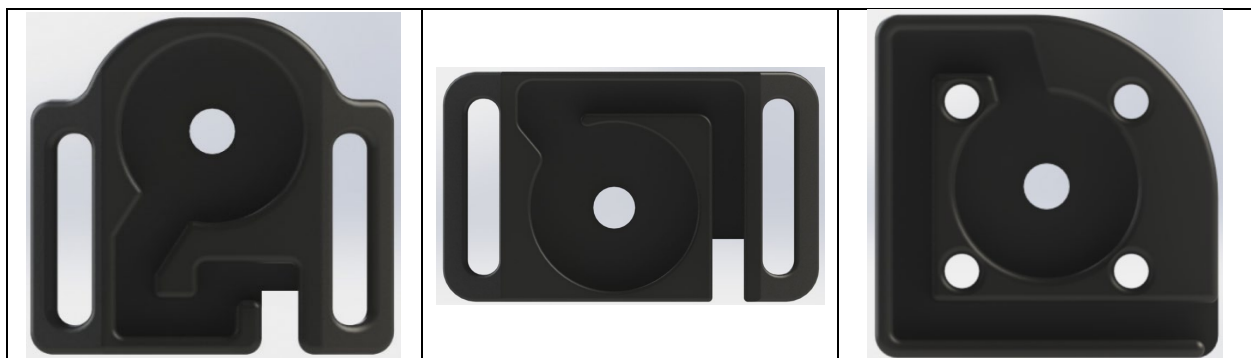


Figure 22: The three motor mount designs are shown, each with included wire routing channels. The fingertip is shown on the left, the finger mount is shown in the center, and the palm is shown on the right. All faces shown are designed to be incident with the user in their respective locations.

On the anterior side of the palm, low-density foam was used to accomplish several objectives. The curvature of the palm varies between people and among hand orientations. The foam helps account for these disparities by compressing to match the curvature of the user's palm. This is especially beneficial for stroke patients that exhibit spasticity, a condition causing muscle stiffness which can occur due to the sensory impairments resulting from a stroke. Finally, the foam increases the comfortability of the device and increases user safety by further enclosing the device wiring.

Attached to the foam are two rigid 3D printed plastic pieces. These plastic pieces both hold hook-and-loop straps for adjustment and press both the foam and motors into the patient's palm. Using two separate pieces allows for the palm to bend and adapt to different patient hand geometries. The design of these palm pieces is ambidextrous, allowing the device to be used on either hand by threading the hook-and-loop straps across the other side of the plastic.

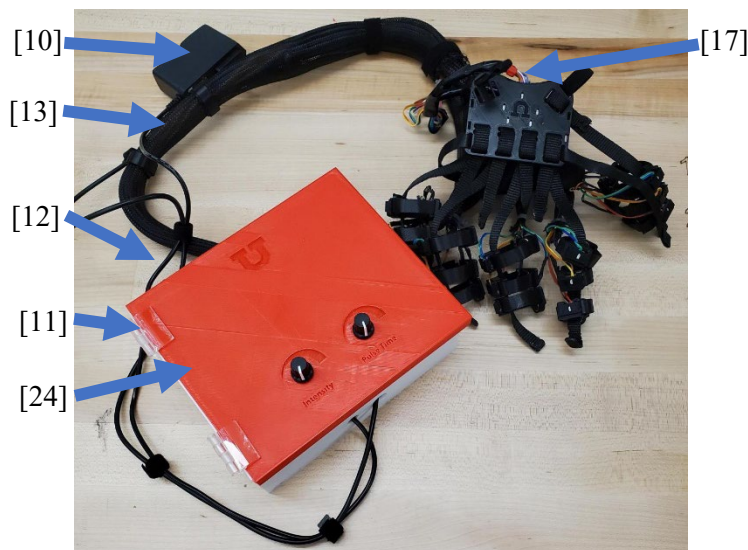


Figure 3: The final prototype design of the wearable haptic device is shown. All electrical components are included within the red and white electrical box.

The palm of the device connects to a third rigid plastic piece which sits on the dorsal side of the hand. This piece contains the LEDs corresponding to the palm motors so the patient can see the visual stimuli while their hand rests on a table. This piece contains the connection points for the finger straps with the option to connect the thumb strap on either the left or right side. For user comfort, foam padding was attached to this rigid piece to pad the back of the hand for user comfortability.

For adjustability, most of the straps on the device are hook-and-loop which enables the user to achieve a custom fit. The hook-and-loop straps are used for the palm of the device, as well as to hold the motor mounts on the fingers. The straps used along the fingers are nylon, because the nylon does not stick to itself nor to the hook-and-loop. By not using hook-and-loop for these straps, the maneuverability of the device, time to put on, usability, and user comfortability were all improved.

The electrical box was 3D printed to contain the circuitry and wiring for user safety. The wiring exiting out of the box was contained in braided wire shielding for safety and general organization.

The data cable used to connect the microcontroller to the computer as well as the power cable are securely mounted into the electrical box to prevent them from being pulled out of the circuit board.

Graphical User Interface

Control of the device in an easy-to-use manner is a critical user need. This was accomplished through a custom Graphic User Interface (GUI). To generate the GUI, the software LabVIEW was chosen. A special toolkit was needed that allowed LabVIEW to be able to write and control the microcontroller's pins.

Within the GUI, instructions were placed on the left side to ensure that the user sees them first. These instructions ensure the program runs as intended and assist the user in learning how to operate the program (Figure 5). Also included are two sliders on the right side of the GUI that allow for the control of the intensity of the vibrational stimulus from the motors and a pulse time slider to control the duration the stimuli are activated. Each slider indicates the active value of the setting to allow for the user to precisely choose the values wanted, or record for later usage. In the top-right of the GUI, a button allows the user to toggle between individual control or group control modes of the device. The button lights up and switches text to explicitly indicate the active mode. The bulk of the GUI is laid out such that each section of the finger and palm are clearly labeled and sectioned off so the user can know explicitly what motor or groups of motors they are activating when pressing the buttons. Figure 4 shows a snapshot of the GUI and the layout of the buttons.

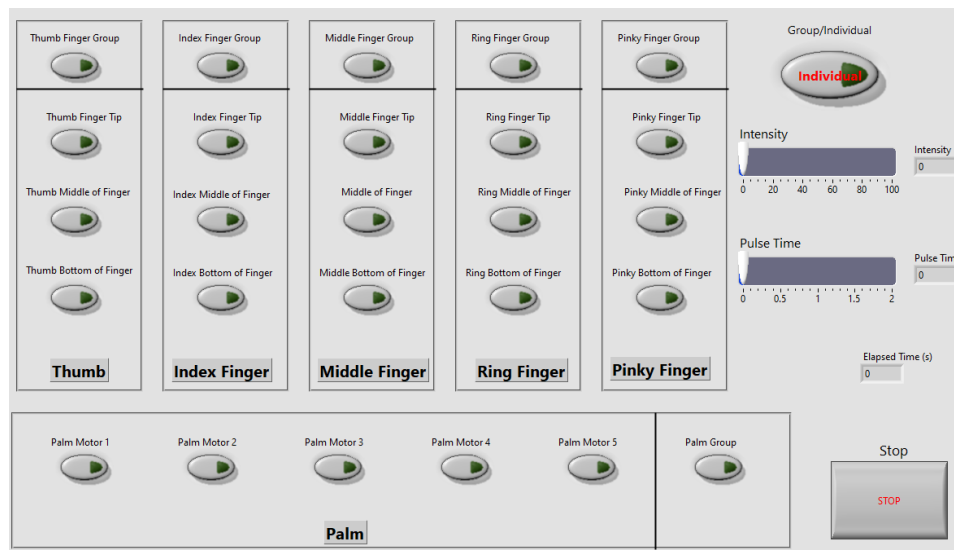


Figure 44: Layout of the Graphical User Interface, each part of the finger and palm are sectioned and include their respective motor-LED pair that is controlled through the buttons. Vibrational intensity and pulse time duration sliders, shown on the right, allow for pulse time and vibrational intensity adjustments. The control mode toggle button, shown in the top right corner, allows users to switch between controlling groups of motors and individual motors.

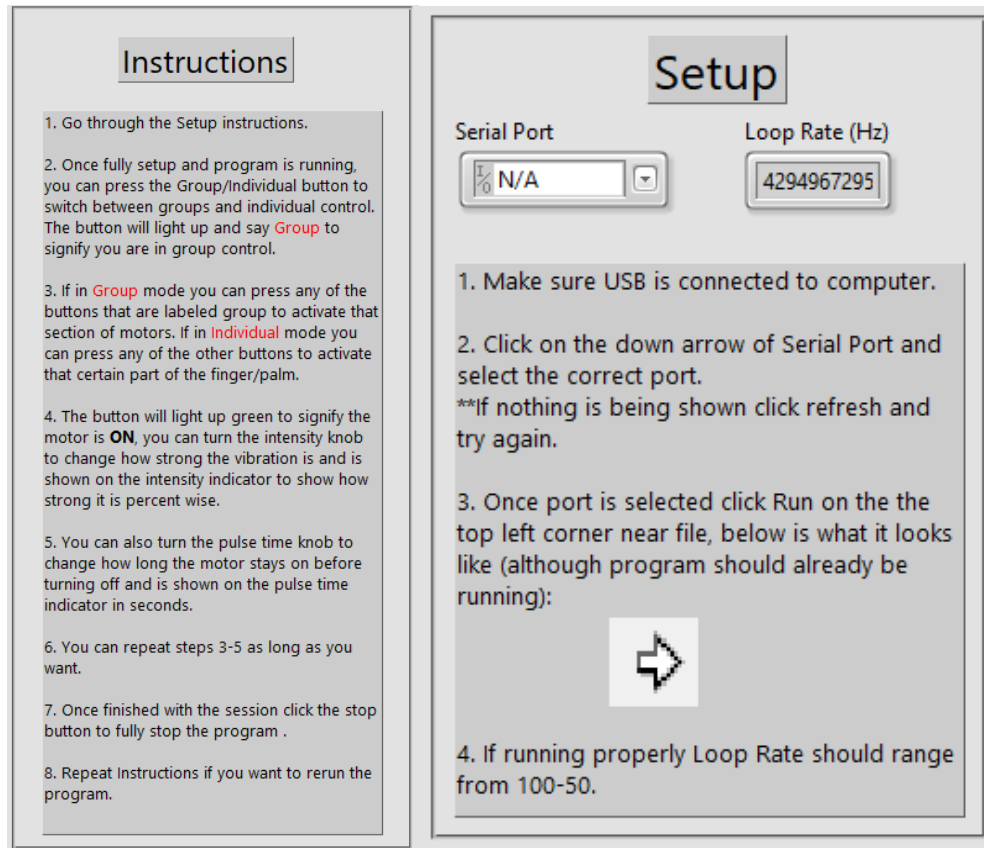


Figure 55: Detailed instructions included in the Graphical User Interface to ensure the program runs as intended and explicitly tell the user how to run the program.

Printed Circuit Board

To hold and secure all the needed wired connections, a custom Printed Circuit Board (PCB) was created to allow for easier wire management, especially for the complex electrical circuit used. As shown in Figure 6, the PCB contains female header connections for motors and LEDs in parallel, such that they would turn on at the same time when given a signal. It also contains connections of other circuitry elements such as the microcontroller, resistors, and transistors. This PCB design allowed the device to use ribbon wire for all connections, which prevents wires from getting tangled during transportation or use. The individual components in the figures are labeled with part numbers corresponding to their placement in the Bill of Materials in Appendix 3.

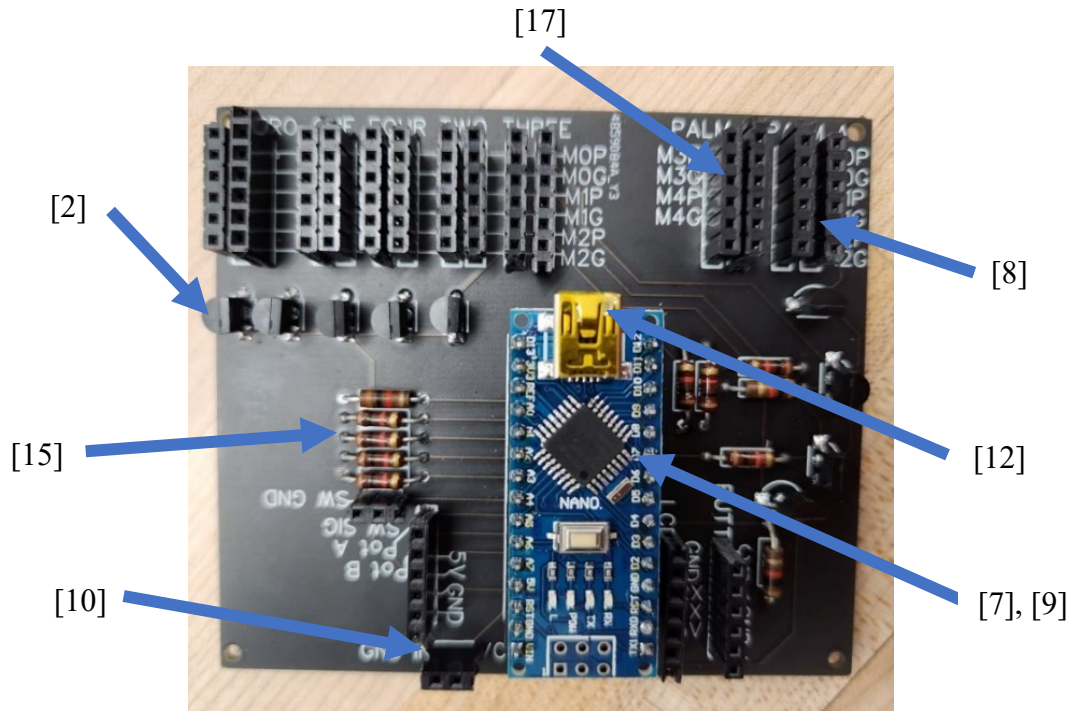


Figure 66: The Printed Circuit Board custom designed using EasyEDA, manufactured by JLCPCB, and assembled by the team. Header pins indicate connection points for the wiring to be made with the wall power adapter and with the motor-LED pairs. The full electrical schematic is in Appendix 5.

Performance Verification

Engineering Analysis

Motor Selection

The wearable device consists of 20 vibrational motors and a major portion of the analysis revolved around deciding which motors would best suit the user's needs. Based on the research completed, explained in Appendix 2, there are 4 major tactile receptors, and each major receptor is sensitive to different frequencies. To guarantee that the vibrational stimuli are felt, a frequency range of 100 – 250 Hz was established to overlap these receptors. From the research, a maximum vibrational amplitude of 1G was established to reduce the risk of patient injury.

Hand Geometry Statistics

One of the user needs was to create a wearable device that is capable of fitting 90% of all adults. This meant that hand measurements needed to be determined for the 5th percentile of women and the 95th percentile of men. Using experimentally determined lengths and typical finger length ratios, the measurements needed were found and compiled, as seen in Appendix 2. Based on these results and by using Equation 1, it was determined that a maximum of 3 motors would fit per finger based on the diameter of the motor.

$$\# \text{ of Possible Motors} = \frac{\text{Shortest Finger Length}}{\text{Motor Diameter} + 2 * \text{Motor Thickness}} \quad (1)$$

Motor Control

Another user need for this device is to be able to control the motors in groups (finger or palm sections) as well as individually. After exploring the applicability of some techniques, the final design utilizes parallel circuits energized via distinct transistor combinations to achieve control of which circuit is active, which can be seen in the custom circuit board shown in Figure 6.

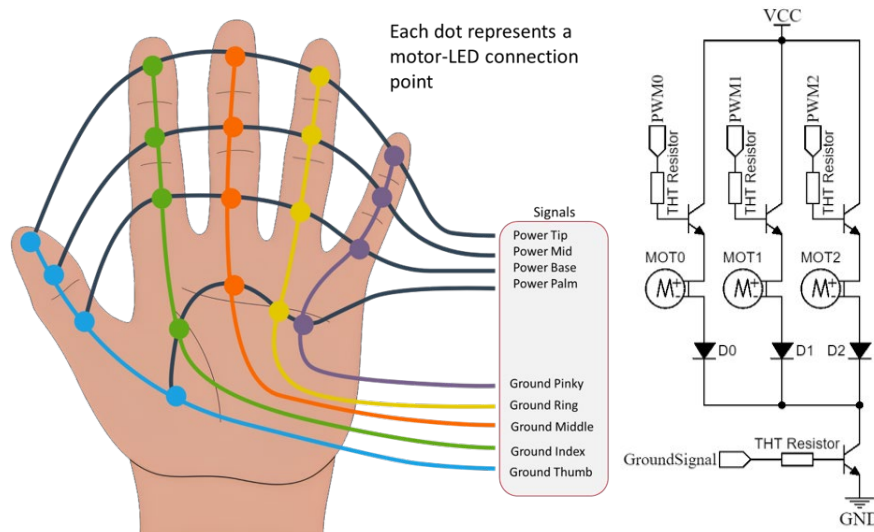


Figure 77: The matrix structure of connections within the device is represented graphically against a backdrop of a hand (left). Colored lines indicate the routing of signals provided by the microcontroller to the device, and each line is labelled within the legend. A sample of the overall circuitry to demonstrate the control methodology for activating motors in groups and individually is also shown (right).

Figure 7 shows a simplified sample of the overall circuitry to explain the control methodology (the full electrical schematic is in Appendix 5). Activating a motor requires two active transistors, a lower one which connects the circuit to ground, and one upper transistor that connects the circuit to power (6V). The transistors are activated by signals sent from the microcontroller. A high voltage activates the ground transistors to saturation, allowing full current to flow through the lower transistors. The frequency and amplitude of motor vibration is adjusted by Pulse Width Modulation (PWM) signals sent to the upper transistors, allowing for continuously varying amounts of current to flow through to the motors. By activating the correct pair of lower and upper transistors, any motor can be activated individually. By cleverly connecting the motors to a common lower transistor (accomplished within the PCB), an entire group can be activated by activating the upper transistors associated with each motor in the group. Thus, electrical control of the motors individually and in groups is achieved.

Testing

Motor Selection

Three types of motors were purchased that generate vibrations using different phenomena for testing:

1. Linear Resonant Actuators (LRAs)
2. Eccentric Rotating Mass actuators (ERMs)
3. Piezo vibration actuators

The motors were tested using a simple test circuit (Figure 8), constructed using a microcontroller, and some potentiometers for real-time adjustments, to determine which motor was the best fit for the device. Each of these actuators achieves vibration via different mechanisms which resulted in different frequency and amplitude characteristics. The microcontroller was programmed to pulse the motors with a pulse duration, a delay between pulses, and frequency/amplitude of the vibrational stimulus specified by the potentiometers. This allowed the team and advisors to gauge the tactile feeling of each motor in relation to the others.

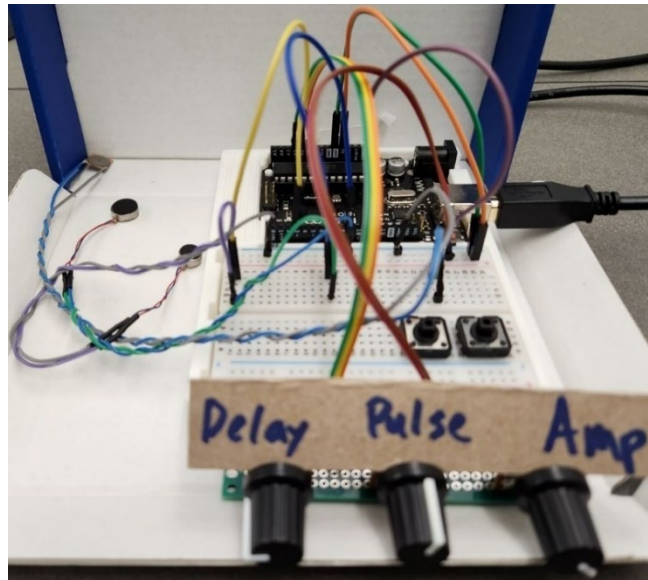


Figure 88: Motor selection test setup showing a microcontroller to control the system and potentiometers for adjustments of vibrational control options for real-time testing of the various vibrational motors.

In collaboration amongst the team members and advisors, the ERMs were chosen as these motors were the most detectable compared to the others that were tested. The piezo vibration actuators required more voltage than could be supplied with supplies on hand and were 15 times more expensive than the others, making them impractical for this project. For the LRAs, the vibrations could be heard, but were difficult to feel due to the small mass of the actuator. 10mm and 8 mm diameter ERMs were purchased for comparison and, due to the larger vibrational mass and surface area, the 10 mm diameter motors were better suited for the project as they provided vibration across a larger area. These motors provide a maximum of 0.75 Gs of acceleration, which is below the maximum limit set in the metrics, see Table 2. Their frequency at 3V correlates to 200 Hz and is also within the project's established range of metrics, see Table 2. By varying the voltage sent to the ERMs the frequency can be varied between 150 Hz and 250 Hz.

Motor Control

To determine whether the designed circuit was capable of actuating motors both individually and in the predetermined groups as required by the project metrics, multiple tests were run:

1. Plugging an individual motor into each motor position on the Printed Circuit Board (PCB) individually verified the continuity of the circuitry and PCB functionality
2. Testing one group at a time ensured that the predefined groups functioned as prescribed

3. Finally, a microcontroller ran a program that cycled through each individual motor followed by the different groups was used to prove that the PCB was capable of full motor control

When testing the control of the motors through manual connection and disconnection of power and ground from the appropriate transistors, actuation of each motor was achieved when connected to the PCB. Additionally, actuation of motor groups was achieved in the same circumstances. Upon implementation of the program to run each motor and group in a sequence, actuation of motors in groups was non-functional, as leakage currents were activating motors that should not have been activated. This was rectified through the inclusion of diodes in line with each motor (explicitly within the motor harnesses) as shown in Figure 9, and all leakage currents were eliminated. The remaining functionality of the device was unaffected by the inclusion of said diodes.

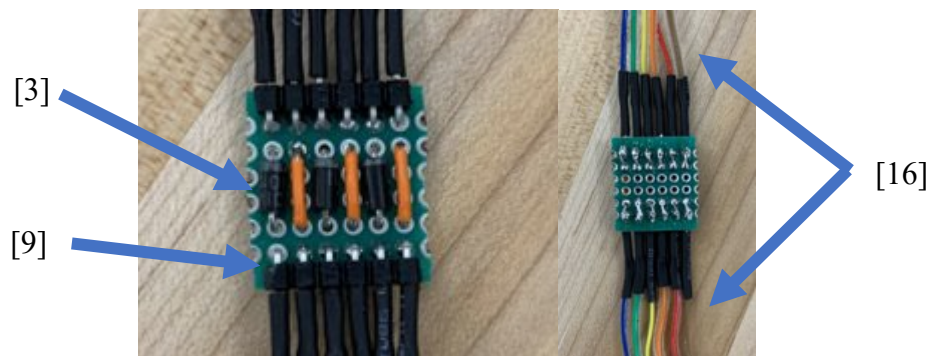


Figure 99: A protoboard created to add diodes into the existing harnesses for the motors. Diodes can be seen in the left image (top face) as black cylinders, and the orange jumpers act as wires to ensure continuity for the return paths from the motors to ground (through the Printed Circuit Board). The right image shows the bottom face of the protoboard.

Upon testing the PCB with a cyclic program, the PCB was proven electrically capable of both individual and group motor control, without leakage current in any locations.

Ease of Use

The associated difficulty for the user to put on the device was an important metric, both with therapist assistance and without. When putting the device on individually, the user had to rely on their opposite hand both to fasten the hook-and-loop straps and position the motors. When assisted, the assistant was able to use both hands to fasten the hook-and-loop straps and position the motors. The associated difficulty was ranked as follows:

1. Difficult
2. Somewhat difficult
3. Neither easy nor difficult
4. Somewhat easy
5. Easy

When recording the amount of time it took to put the glove on (both assisted and unassisted), the established metrics were only barely achieved. The results are shown in Table 2. The achieved application times were 2.8 minutes assisted and 4.5 minutes unassisted. The ease-of-use metric was also achieved, as the trials ranked at 3, neither easy nor difficult. Although faster application

times are desirable, there was a tradeoff between full adjustability and ease-of-use. It was determined that adjustability expands the potential use of the device to more patients and was thus deemed of greater importance than the device being extremely quick and easy to put on.

Device Fit

After completion of glove construction, the metrics for finger fit (metrics 9 and 10 in Table 2 and Appendix 2) were achieved. As an informal verification, each member of the team and the advisors also tried on the device to verify the adjustability and fit of the finger and palm straps were satisfactory. The prototype fit is shown in Figure 10, on a 3D printed hand, for demonstration purposes only.



Figure 10: The full prototype device demonstrated on a 3D printed hand for demonstration purposes. The left image is of the dorsal side of the hand, and the right picture is of the palmar side of the hand.

Project Planning

For the schedule of the project, a timeline was established at the beginning of each semester with deadlines for each milestone, as shown in Appendix 8. Tasks in the beginning included preliminary designs, testing, schematics, and ordering of necessary electronic components. Further milestones included assembly of the device, programming, and further testing. The final milestone was Design Day which involved presenting the final, completed device to peers, faculty, guests, and engineers from industry. As the project progressed, part delays, scheduling conflicts, and scope changes led to some deadline delays, which are shown in Appendix 8. Holistically, all project milestones were completed. Despite deadline adjustments, the overall project was completed on time and successfully.

Budget

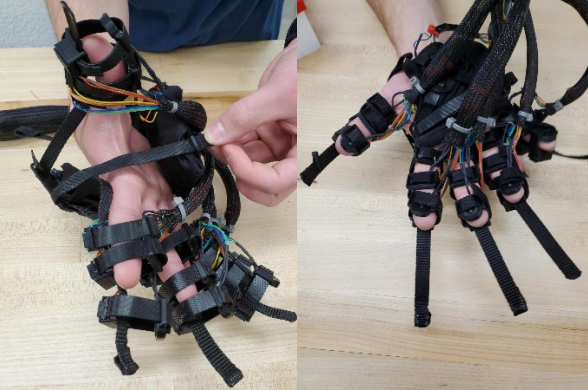
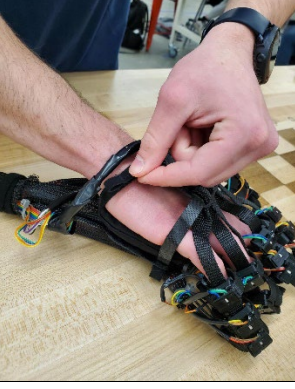
The overall budget for this project was constrained to \$500. The project used a total of \$408.91 and came in under budget. During research and development, \$115.10 was spent that did not end up within the final prototype. This included the motors that were tested and not chosen for the final design, electronic components that were eliminated in favor of superior designs, and various other

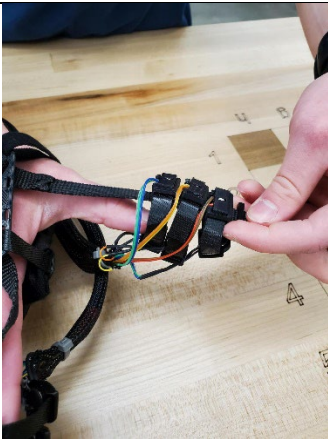
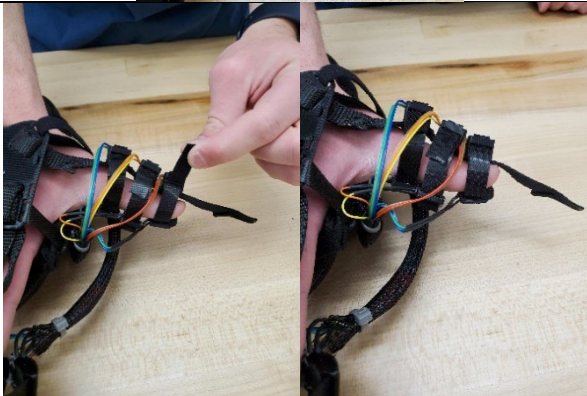
prototyping conveniences. The breakdown of the total budget is included in Appendix 7. The Bill of Materials for the device is included in Appendix 3 and includes the vendors each part was sourced from along with the associated part number.

Conclusion

The project was an overall success and was completed on time, and under budget. The resulting prototype is functional and provides vibrational and visual stimuli to the affected hand. The stimuli are adjustable and can be activated both groupwise and individually through a Graphical User Interface (GUI). The GUI can be operated on any personal computer without downloading any proprietary software, and interfaces with the device via the incorporated USB mini-B cable. The device is lightweight to reduce user fatigue. The device is adjustable to the specific conditions of the user, including being reversible to fit either hand and is adjustable to fit comfortably on the palm, while also allowing fine placement of the motor-LED pairs along each finger. The device is not overly complicated to put on nor to take off and is safe for the user and clinician. All necessary paperwork has been submitted to the Institutional Review Board (IRB), and the device has been deemed safe to be used and tested with stroke patients. The creation of this novel prototype represents the beginning of new therapies and techniques for stroke rehabilitation. The prototype and the project will be handed off to Dr. Mohammed Sbaji, a clinical neurophysiologist, who will begin developing the procedures and protocol associated with this device, for the betterment of medicine, and increased recovery opportunity for stroke patients.

Appendix 1: Device Adjustment Instructions

	1. Place affected hand into device, with the wearable fingers placed loosely around each finger. Ensure the LEDs of the palm are on the back of the hand. The LEDs are in the rigid rectangular piece with the debossed U present.
	2. Wrap the hook-and-loop strap between the thumb and index finger around the palm and insert through the coordinating loop in the rigid plastic on the palm side of the hand. Pull tight and press flat across the palm to ensure a snug connection of the two hook-and-loop straps to lock it in place. Make sure that the fit is snug and comfortable.
	3. Wrap the hook-and-loop strap below the thumb around the thumb and insert through the coordinating loop in the rigid plastic on the back of the hand. Pull until the desired snug yet comfortable fit is achieved and fold the hook-and-loop strap back on itself to lock it in place.
	4. Repeat step 3 for the straps on the pinky side of the hand, tightening the palm portion of the device to the hand and ensuring a snug but comfortable fit throughout the entirety of the palm.

	5. Starting with the thumb, lift the finger motors up the nylon strap and slide the rigid plastic mounts toward the palm, placing the motors in the center of each finger segment.
	6. Pull the hook-and-loop straps so that the motor in each segment sits comfortably snug against the skin.
	7. Repeat steps 5 and 6 with all remaining fingers. The nylon straps that extend beyond the fingertips can be tucked away, by folding them over the finger towards the palm, or left out as preferred.

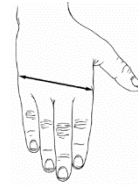
Appendix 2: Engineering Analyses

Statistical Hand Sizes

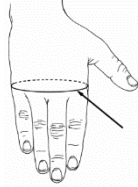
The following tables show the average hand measurements for the 5th and 95th percentile for both males and females [1]. Index and ring finger length calculations are sourced from experimental data [2] and using Equation 3 while the rest of the fingers are calculated by multiplying experimentally determined finger length ratios [3, 4] with the corresponding determined finger length. The middle finger length was averaged between the length calculated using the index-middle finger ratio and the ring-middle ratio.

$$5\text{th, } 95\text{th Percentile} = \text{Mean Finger Length} \pm 2 * \text{St. Dev} \quad (3)$$

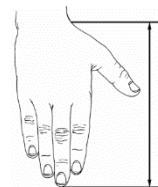
<i>Hand Breadth</i>						
Percentile:	5%		50%		95%	
Units:	cm	in	cm	in	cm	in
Female:	7.34	2.89	7.93	3.12	8.56	3.37
Male:	8.36	3.29	9.03	3.56	9.76	3.84



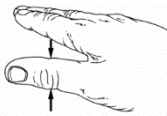
<i>Hand Circumference</i>						
Percentile:	5%		50%		95%	
Units:	cm	in	cm	in	cm	in
Female:	17.25	6.79	18.6	7.32	20.03	7.88
Male:	19.85	7.81	21.34	8.4	23.03	9.07



<i>Hand Length</i>						
Percentile:	5%		50%		95%	
Units:	cm	in	cm	in	cm	in
Female:	16.5	6.5	18.02	7.09	19.69	7.75
Male:	17.87	7.04	19.33	7.61	21.06	8.29



<i>Thumb Breadth</i>						
Percentile:	5%		50%		95%	
Units:	cm	in	cm	in	cm	in
Female:	1.86	0.73	2.06	0.81	2.29	0.9
Male:	2.19	0.86	2.41	0.95	2.65	1.04



Finger	Mean	St. Dev.
Male Index	73.15	4.82
Male Ring	73.68	4.66
Female Index	67.78	4.03
Female Ring	68.18	4.63

Finger Ratio	Male	Female
Middle:Index	1.108	1.106
Middle:Ring	1.072	1.081
Pinky:Middle	0.766	0.744
Thumb:Index	0.690	0.710

Finger Lengths							
Percentile:		5%		50%		95%	
Units:		cm	in	cm	in	cm	in
Male	Index:	6.35	2.50	7.32	2.88	8.28	3.26
	Middle:	6.97	2.74	8.00	3.15	9.03	3.56
	Ring:	6.44	2.54	7.37	2.90	8.30	3.27
	Pinky:	5.34	2.10	6.13	2.41	6.92	2.72
	Thumb:	4.38	1.72	5.05	1.99	5.71	2.25
Female	Index:	5.97	2.35	6.78	2.67	7.58	2.98
	Middle:	6.49	2.56	7.43	2.93	8.38	3.30
	Ring:	5.89	2.32	6.82	2.69	7.74	3.05
	Pinky:	4.83	1.90	5.53	2.18	6.24	2.46
	Thumb:	4.24	1.67	4.81	1.89	5.38	2.12

Sources:

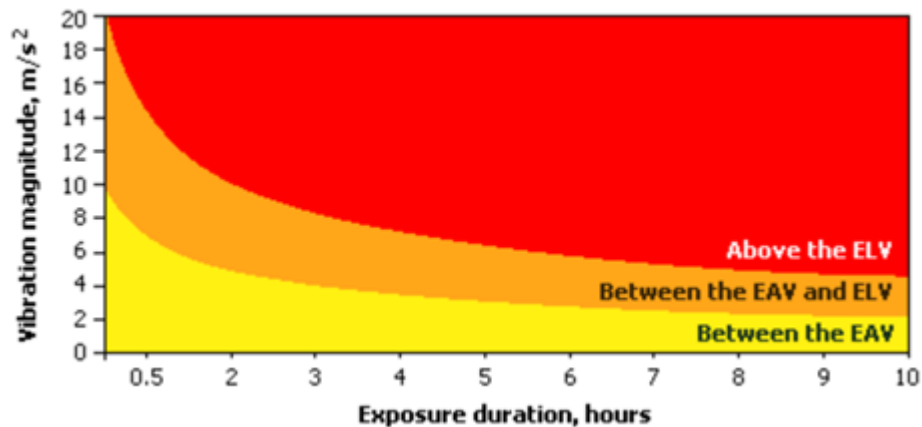
- [1] Ergonomics Center's Anthropometric Data published in 2006.
- [2] Akhlaghi, Mitra, Ardeshtir Sheikhazadi, Fatemeh Khazaie, Shima Rafiee, and Mohammad Hossein Sadeghian. "The Role of Index and Ring Fingers in Gender Identification and Height Estimation." International Journal of Medical Toxicology and Forensic Medicine 9.2 (2019): 75-82. Web.
- [3] Voracek, Martin, and Denise Offenmuller. "Digit Ratios (2D:4D and Other) and Relative Thumb Length: A Test of Developmental Stability."
- [4] Voracek, Martin. "Sex and Side Differences in Relative Thumb Length." The Journal of Hand Surgery (American Ed.) 31.2 (2006): 326-27. Web.

Frequency and Amplitude Specifications Within Human Hands

The following table and chart show the susceptible frequency and amplitude ranges that allow for a safe vibrational stimulus to be felt. Each of the 4 major tactile receptors are highly sensitive with their own receptive field size. Fast-responding receptors will respond to stimuli briefly while slow-responding receptors will respond to stimuli if it is still going [5]. Each of these receptors has their own frequency range and so a frequency range was chosen that could be achieved by a haptic actuator but still stimulates most of the receptors, 100-400 Hz [6].

Receptor	Adaption Speed	Receptive Field Size	Frequency Range (Hz)
Meissner	Fast	Small	30 - 50
Pacinian	Fast	Large	250 - 350
Merkel	Slow	Small	0 - 5
Ruffini	Slow	Large	100 - 500

For vibrations, the amplitude can be measured by relation to the gravitational acceleration, g. This allows for correlation to a force that is applied to the object. From research done, about 1-0.5 g is reasonable for short durations but quickly drops the longer the exposure to the vibration is. This can be easily seen from a graph provided by the Health and Safety Executive, HSE. The orange region shows the threshold values for amplitude so that vibrations can be felt. Anything higher than the region can cause internal damage and anything lower than the region could lead to the vibration not being felt [7].



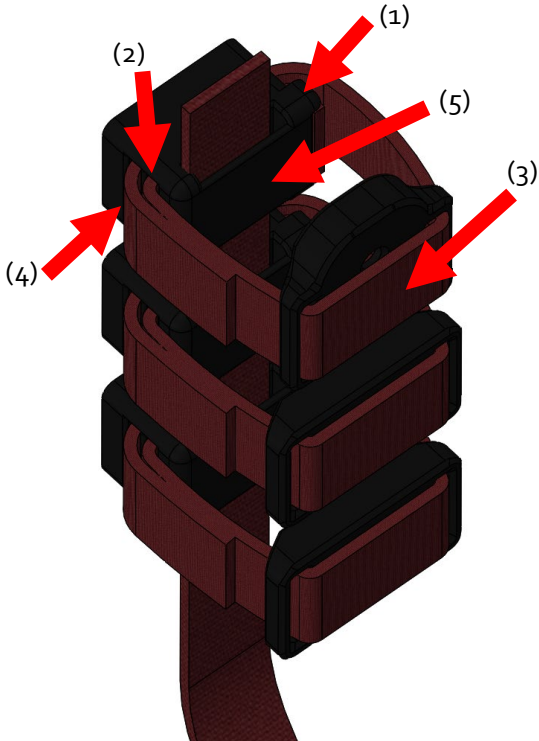
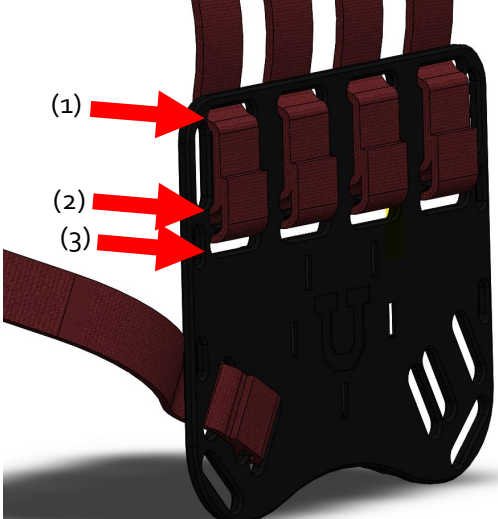
Sources:

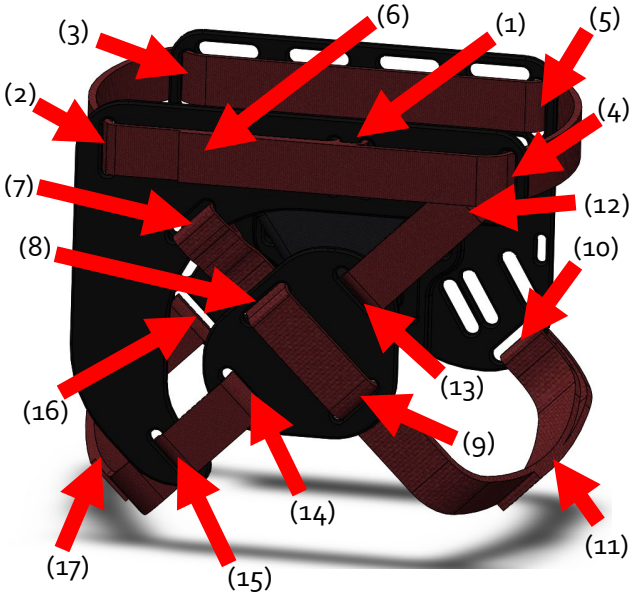
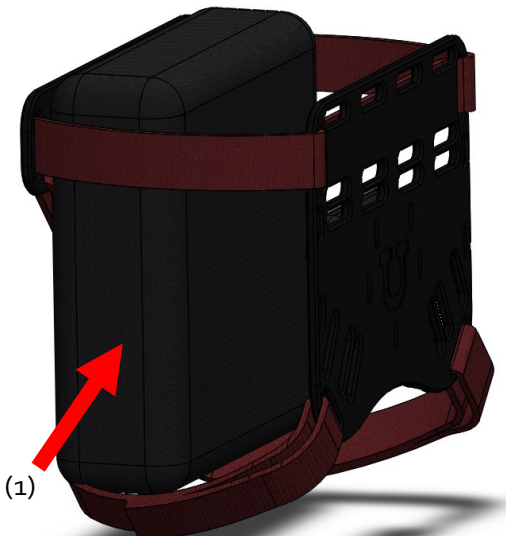
- [5] Forouharmajd, Farhad et al. "Hand-arm Vibration Effects on Performance, Tactile Acuity, and Temperature of Hand." Journal of medical signals and sensors vol. 7,4 (2017): 252-260.
- [6] Moshourab, Rabih et al. "Measurement of Vibration Detection Threshold and Tactile Spatial Acuity in Human Subjects." Journal of visualized experiments : JoVE ,115 52966. 1 Sep. 2016, doi:10.3791/52966
- [7] Centers for Disease Control and Prevention. (2014, June 6). Occupational exposure to hand-arm vibration (89-106). Centers for Disease Control and Prevention. Retrieved March 8, 2023, from <https://www.cdc.gov/niosh/docs/89-106/default.html>

Appendix 3: Bill of Materials

Item #	Item	Descriptions	Vendor	Part ID	Quantity
1.	Motors	Shaftless Vibration Motor 10x3, Eccentric Rotating Mass	Pololu	Po1636	20
2.	Transistors	BJT - NPN Transistors	Digikey	MPSA06RA	8
3.	Diodes	Rectifier Diodes	Amazon	B07Q3HBM63	20
4.	LEDs	LED Sequins	Amazon	B07F23ZN2B	20
5.	Velcro	1 cm Velcro roll (quantity in yards)	Amazon	B095PNTVJ4	1
6.	Fabric	Black fabric: 95% polyester, 5% Lycra Spandex (qty in yds)	Amazon	B00LXCOLKE	1
7.	Arduino Nano	Elegoo Arduino Nano	Amazon	EL-CB-005	1
8.	Female Headers	Female header pins for PCB connections	Amazon	B07SNR8QL6	1
9.	Male Headers	10 pcs male header pins, 40 header strips	Amazon	B07PKKY8BX	1
10.	Power Supply	6V, 2A power supply for motor power	Amazon	B089Q9ZL2C	1
11.	Hinges	Clear adhesive mini hinges	Amazon	B0BG81VQMF	1
12.	Arduino Cable	Arduino cable to connect to GUI	Amazon	B0BF9NPGW3	1
13.	Wire Shielding	Braided wire shielding (quantity in packs)	Amazon	B09B9NCWX6	1
14.	Nylon Webbing Strap	3/8" Nylon Webbing for finger straps	Amazon	B09QJ4LDRT	1
15.	Resistors	1 K Ω resistors	-	-	12
16.	Ribbon Cable	3 ft long ribbon cable segments per hand section	-	-	6
17.	PCB	Custom PCB	JLCPCB	-	1
18.	Fingertip Mounts	3D printed fingertip motor mounts	Custom	-	5
19.	Finger Mounts	3D printed finger motor mounts	Custom	-	10
20.	Palm Mounts	3D printed palm motor mounts	Custom	-	5
21.	LED Holders	3D printed LED holders	Custom	-	15
22.	Finger Adjustment Sliders	3D printed sliders for finger strap adjustments	Custom	-	15
23.	Hand Pieces	3D printed palm, back of hand, and corresponding padding	Custom	-	1
24.	Electrical Box	3D printed box to house PCB and wiring	Custom	-	1

Appendix 4: Assembly Diagram

	Motor Strap Assembly: 1. Insert hook-and-loop strap into the holding slot, (1). 2. Then pass the hook-and-loop strap through the passage, (2). 3. Pass the strap through the motor such that the hook-and-loop strap goes across the back of the motor mount, (3). 4. Pass the hook-and-loop strap goes through the final passage in the slider, (4). 5. Finally, pass the finger strap through the vertical slot on the slider, (5). 6. Repeat steps 1-5 three times per finger.
	Finger/Back of Hand Assembly: 1. Pass the finger strap through slots (1), (2), and (3) on the plastic piece on the back of the hand in order. 2. Pull the strap through far enough to achieve the desired finger length. 3. The pulled through strap can then be put back through the slot closest to the edge of the plastic piece (1) to hold it in place. 4. Repeat steps 1-5 for all five fingers, placing the thumb on the correct side corresponding to patient handedness (4).

	Palm/Back of Hand Assembly: 1. The knuckle strap is attached at slot (1). 2. Pass the knuckle strap through slots (2), (3), (4), and (5) in order. 3. Attach the hook-and-loop strap along the front of the knuckles (6). 4. The thumb side strap is attached at slot (7). 5. Pass the thumb side strap through slots (8), (9), and (10) in order. 6. Attach the hook-and-loop strap under the thumb (11). 7. The thumb side strap is attached at slot (12). 8. Pass the thumb side strap through slots (13), (14), (15), and (16) in order. 9. Attach the hook-and-loop strap under the thumb (17).
	Palm Motor Assembly: 1. Place the foam/motor assembly on the back of the palm pieces and secure it using the hook-and-loop straps (1).

Appendix 5: Electrical Schematics

The electrical schematic and resulting ECAD drawing of the custom PCB were generated using EasyEDA. The electrical schematic, shown in Figure 11, maps the connections the circuit board is required to make. Due to the matrix nature of the circuit's connections, all semblance of the actual circuit was lost in favor of making the schematic readable. This resulted in nearly all connections being formed via netlabels, and not drawn wire connections. These netlabels come in pairs, and a connection is made via placing each netlabel at the locations the connections need to terminate. For example, in the top left box of the schematic, marked “Wallwart Power to Transistors, ard”, a netlabel for TipPower makes the connection to the collector of the transmitter associated with powering the tips of each finger, shown in the box located in the left side, middle of the schematic marked “FingerTip Power”.

The ECAD drawing of the PCB, shown in Figure 12, illustrates the resulting layout of components, and the traces embedded within the PCB to achieve the necessary connections. Table 5 explains the significance of the colors utilized within the drawing.

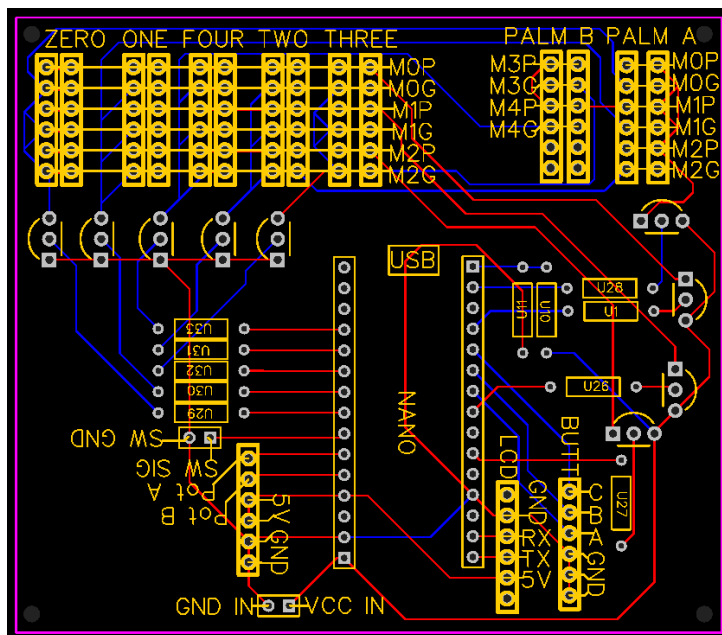


Table 5: Legend for the colors used in the ECAD drawing of the custom designed PCB, and their significance in the design.

PCB Color	Meaning	Layer
Yellow	Silkscreen	Top
Red	Traces	Top
Blue	Traces	Bottom
Purple	PCB outline	-
Silver	Solder pads	-

Figure 11: The ECAD drawing of the PCB shows the resulting layout of components, and the traces embedded within the PCB to achieve the necessary connections.

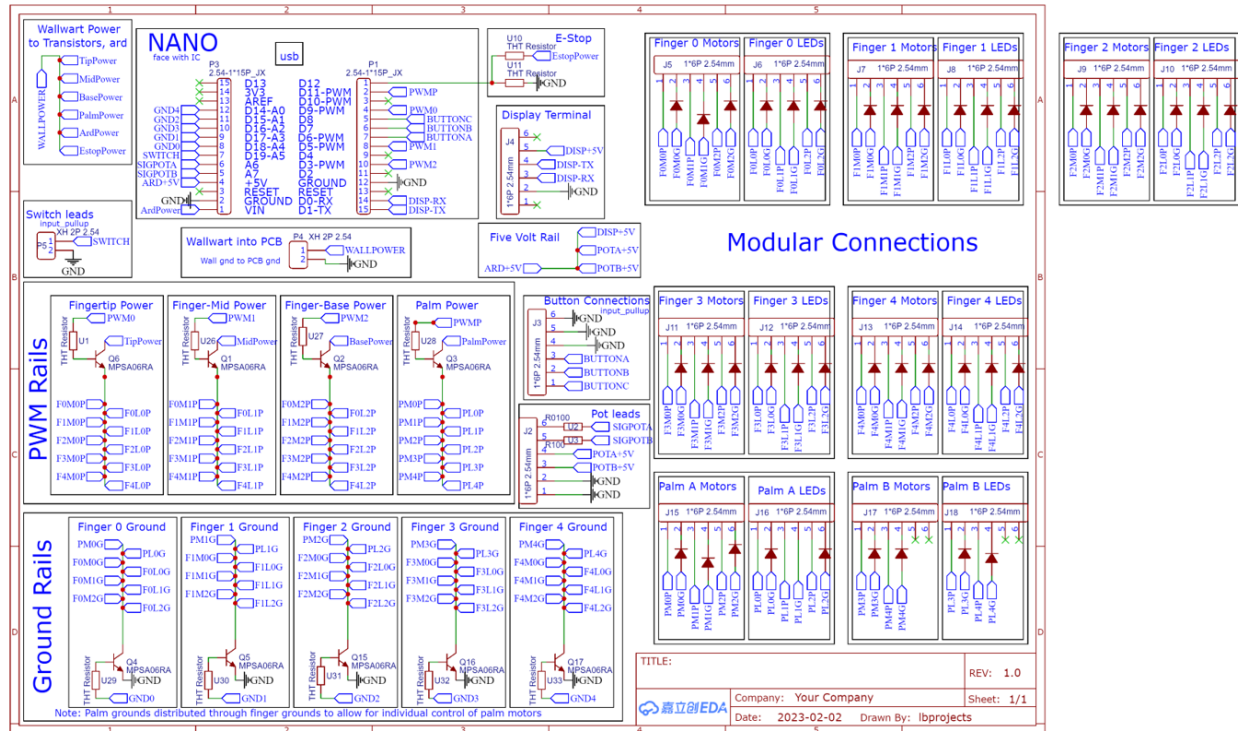
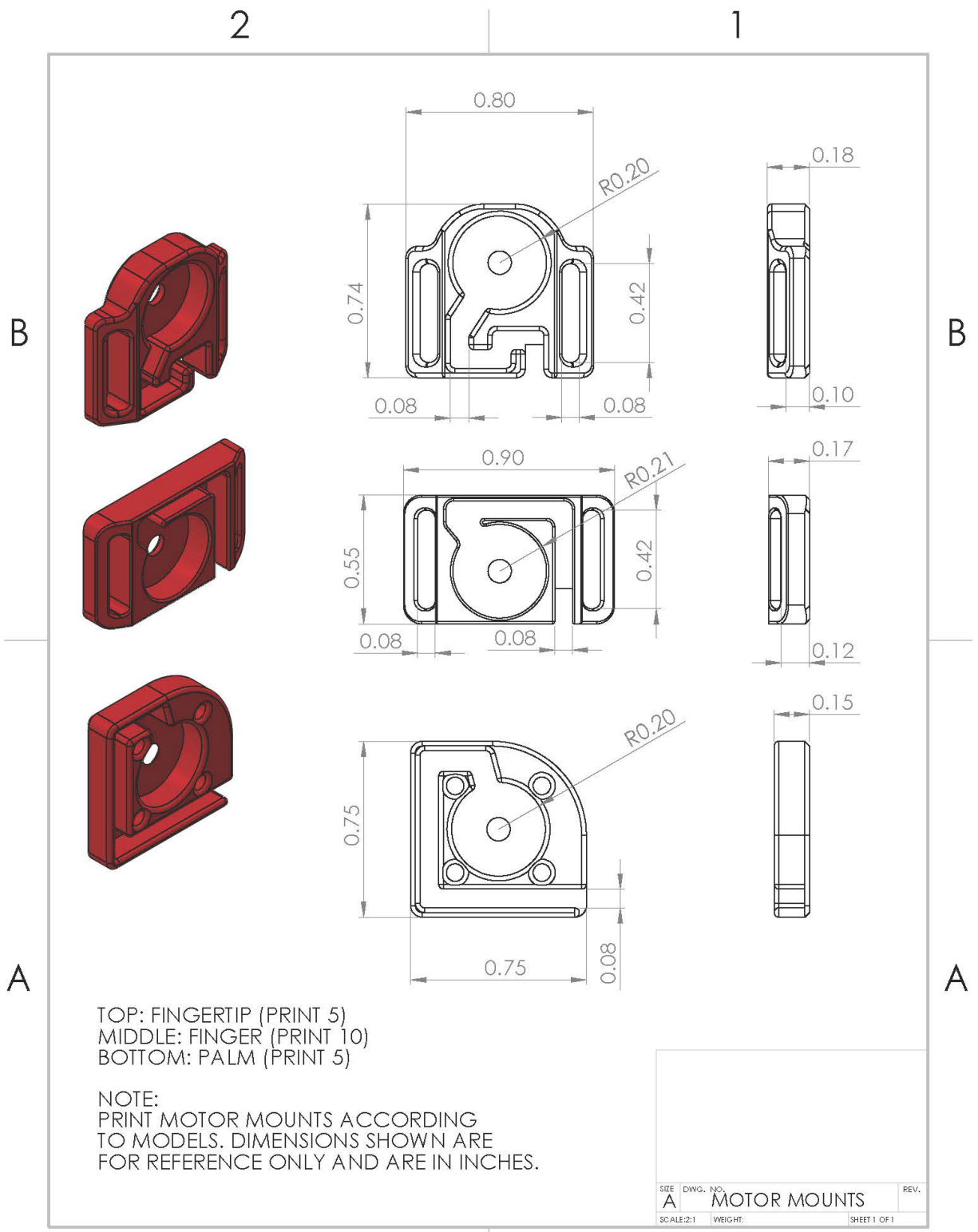


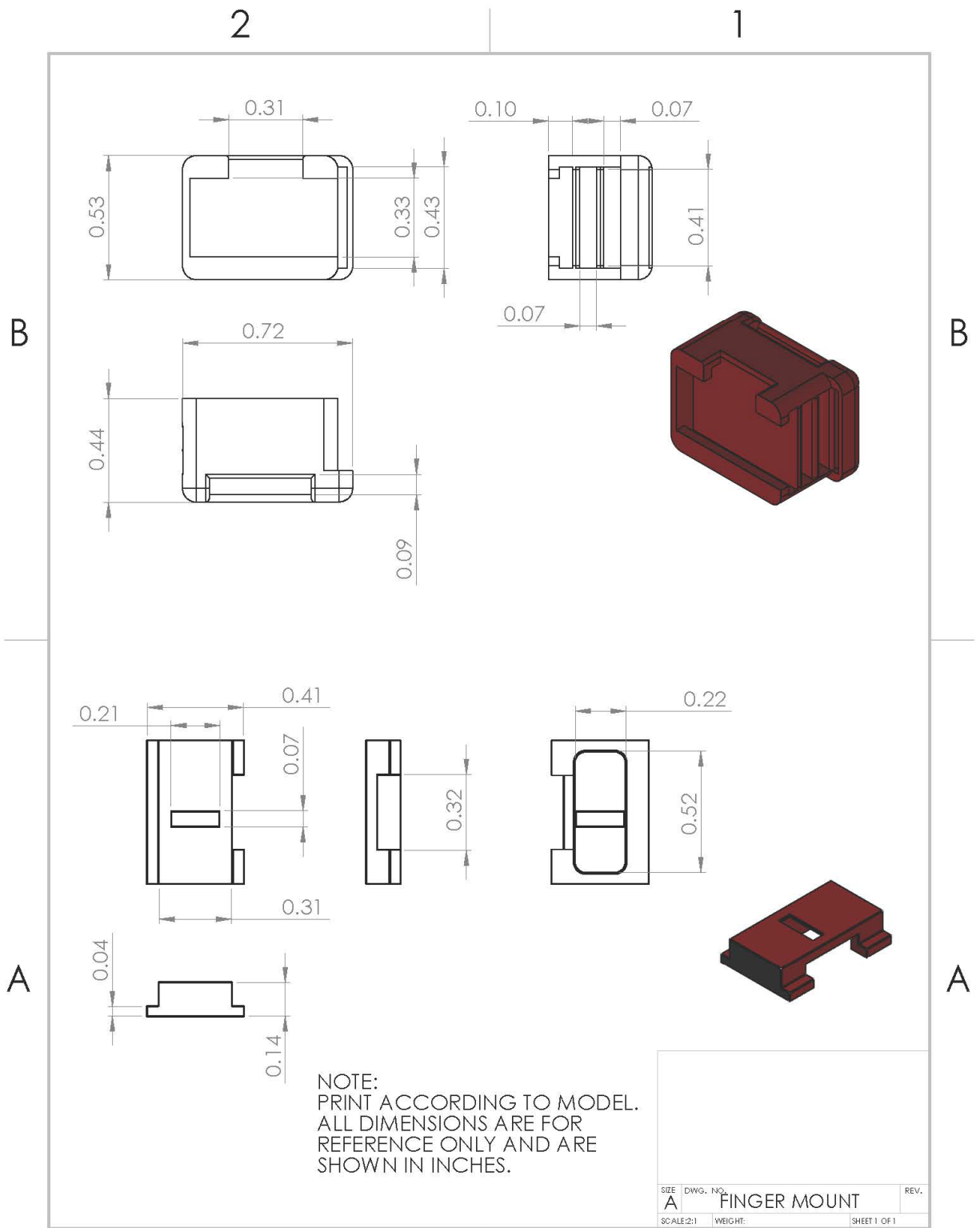
Figure 12: The full electrical schematic utilized within the device. All electrical connections are included, though many utilize netlabels instead of drawn wires to form connections. Netlabels come in pairs, and each netlabel indicates the termination of the connection in consideration.

Appendix 6: Drawings

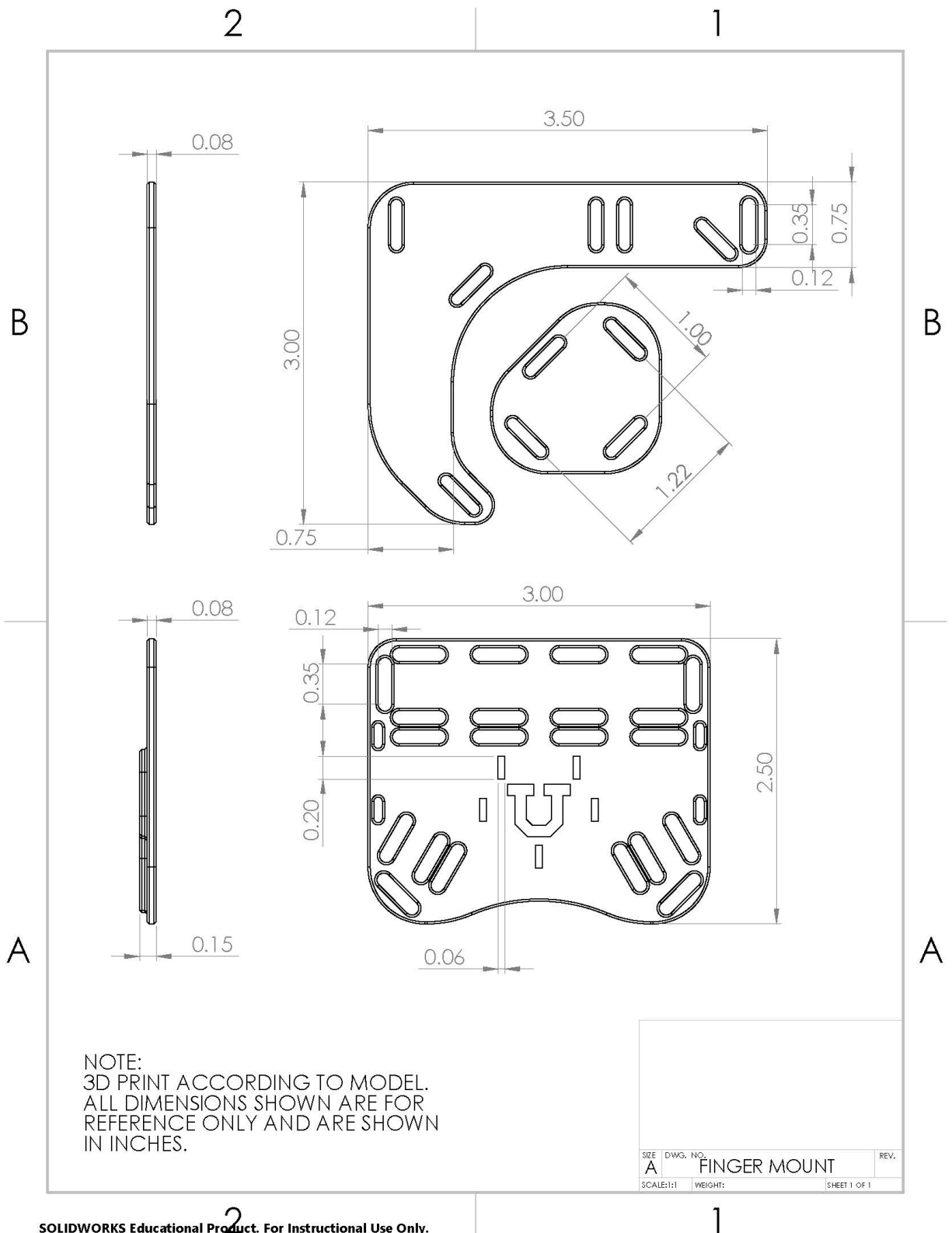
Motor Mounts



LED Slider



Palm and Back of Hand



Palm Portion and Velcro Straps

2	1																
PALM: CUT 2 1/2" SEAM BACK: CUT 2 1/4" SEAM	INSTRUCTIONS: CUT OUT PATTERN PIECES TO CORRECT SIZE SEW WITH RIGHT SIDES TOGETHER, LEAVING SPACE TO TURN RIGHT SIDE OUT GLUE MOTORS IN MOTOR MOUNTS ONTO LARGE FOAM BLOCK STUFF LARGE FOAM BLOCK INTO PALM PIECE STUFF SMALL FOAM BLOCK INTO BACK PIECE SEW BACK PIECE COMPLETELY CLOSED																
PALM FOAM: CUT 1 BACK FOAM: CUT 1	PALM FOAM: CUT 1 BACK FOAM: CUT 1																
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="2" rowspan="2"> DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL: ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ± MATERIAL FINISH </td> <td colspan="2"> NAME DATE </td> </tr> <tr> <td colspan="2"> DRAWN CHECKED ENG APPR. MFG APPR. Q.A. COMMENTS: </td> </tr> <tr> <td colspan="2"> NEXT ASSY USED ON </td> <td colspan="2" rowspan="2"> Hand Parts Sewing Pattern </td> </tr> <tr> <td colspan="2"> APPLICATION DO NOT SCALE DRAWING </td> </tr> <tr> <td colspan="2"> SIZE A DWG. NO. REV. </td> <td colspan="2"> SCALE: 1:1 WEIGHT: SHEET 1 OF 1 </td> </tr> </table>		DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL: ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ± MATERIAL FINISH		NAME DATE 		DRAWN CHECKED ENG APPR. MFG APPR. Q.A. COMMENTS: 		NEXT ASSY USED ON 		Hand Parts Sewing Pattern		APPLICATION DO NOT SCALE DRAWING		SIZE A DWG. NO. REV. 		SCALE: 1:1 WEIGHT: SHEET 1 OF 1	
DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL: ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ± MATERIAL FINISH				NAME DATE 													
		DRAWN CHECKED ENG APPR. MFG APPR. Q.A. COMMENTS: 															
NEXT ASSY USED ON 		Hand Parts Sewing Pattern															
APPLICATION DO NOT SCALE DRAWING																	
SIZE A DWG. NO. REV. 		SCALE: 1:1 WEIGHT: SHEET 1 OF 1															

Appendix 7: Budget

Item Name	Unit Cost*	Shipping	Quantity	Item Total:
Research & Development - Not Included in Final Device				
ELEGOO Arduino Mega	\$19.99	\$0.00	1	\$19.99
Shaftless Vibration Actuator (8mm)	\$4.95	\$4.45	1	\$9.40
Piezoelectric Vibrational Actuator	\$13.59	\$4.99	1	\$18.58
Linear Resonant Actuator	\$3.32	\$5.00	1	\$8.32
LEDs (100 pcs)	\$4.99	\$0.00	1	\$4.99
Switches	\$0.00	\$0.00	1	\$0.00
Multiplexers (3 pcs)	\$8.99	\$0.00	1	\$8.99
Dupont Jumpers	\$7.98	\$0.00	1	\$7.98
NO SPST Momentary Buttons	\$10.99	\$0.00	1	\$10.99
SPST Rocker Switches	\$5.99	\$0.00	1	\$5.99
LCD Display	\$12.89	\$0.00	1	\$12.89
Mini Hinges	\$6.98	\$0.00	1	\$6.98
R&D Subtotal:	\$115.10			
Final Device Budget				
Shaftless Vibration Actuator (10mm)	\$4.19	\$6.45	34	\$148.91
Wiring, fasteners, etc. **	\$0.00	\$0.00	1	\$0.00
3D printing filament **	\$0.00	\$0.00	1	\$0.00
Potentiometers **	\$0.00	\$0.00	4	\$0.00
Rectifier Diodes (125 pcs)	\$5.99	\$0.00	1	\$5.99
BJT - NPN Transistors	\$0.39	\$4.99	10	\$8.89
Male Headers	\$4.99	\$0.00	1	\$4.99
Female Headers	\$1.72	\$0.00	2	\$3.44
Fabric: Poly/Lycra	\$9.99	\$0.00	1	\$9.99
LED Sequins (20 pcs)	\$9.75	\$0.00	3	\$29.25
Velcro Roll - 1cm	\$10.99	\$0.00	1	\$10.99
Articulating Hand	\$9.91	\$0.00	1	\$9.91
ELEGOO Arduino Nano (3 pcs)	\$22.99	\$0.00	1	\$22.99
Plug-In DC Power Supply (6V - 2A)	\$7.99	\$0.00	3	\$23.97
Custom PCB (5 pcs) **	\$2.00	\$19.64	1	\$21.64
Acrylic Hinges (12 pcs)	\$6.98	\$0.00	1	\$6.98
Nylon Webbing Strap - 3/8"	\$9.95	\$0.00	1	\$9.95
Arduino Nano Cable - USB 2.0 to USB Mini-B	\$8.99	\$0.00	1	\$8.99
Cable Shielding	\$14.49	\$0.00	1	\$14.49
Final Device Subtotal:	\$341.37			
Project Total: \$456.47				

** Indicates parts on-hand or donated

Appendix 8: Project Schedule

Project Milestones			
Task	Due Date	Revised Date	Status
Program Flow	10/31/2022	-	●
State Diagram	11/4/2022	-	●
Actuator Apparatus Design	11/4/2022	-	●
Control Panel Design	11/4/2022	-	●
Actuator Selection	11/16/2022	-	●
Electronics Purchasing	11/16/2022	-	●
Actuator Apparatus Modelling	11/16/2022	-	●
Control Panel Modeling	11/16/2022	-	●
Actuator Apparatus Construction	11/18/2022	11/30/2022	●
Control Panel Construction	11/18/2022	11/30/2022	●
Electrical Schematics	11/23/2022	-	●
Actuator Testing	11/23/2022	-	●
CITI Training	11/23/2022	1/9/2023	●
Programming	11/28/2022	11/30/2022	●
Proof-of-Concept Testing	12/2/2022	-	●
Determine wearable device design	1/26/2023	-	●
Achieve individual motor control	2/9/2023	-	●
Determine and purchase visual display	2/14/2023	-	●
Finalize wearable device circuitry and wire management	2/21/2023	-	●
Assemble wearable device	2/28/2023	-	●
Create and order PCB	2/28/2023	-	●
Order all control panel components	2/28/2023	-	●
Complete midterm prototype program	2/28/2023	-	●
Midterm Prototype	2/28/2023	-	●
Finalize control panel design	2/28/2023	3/28/2023	●
Assemble control panel	3/16/2023	4/14/2023	●
Complete wearable device refinement	3/23/2023	4/14/2023	●
Complete control panel programming	3/23/2023	4/14/2023	●
Complete GUI Program	3/31/2023	4/14/2023	●
Test Device	4/4/2023	4/18/2023	●
Design Day	4/20/2023	-	●